



STUDENAC
MARKET

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Self-service checkout potential assesment

April 2026



Retail customer journeys differ based on assistance provided

Type of journey, % of retail trade

Standard
80-85%



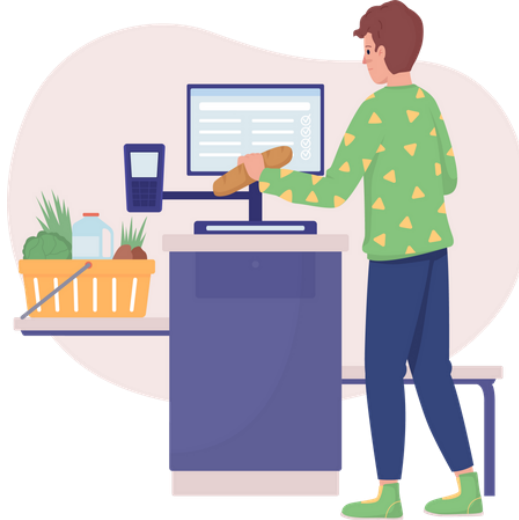
Staff – operated checkout

Personal service, assistance available, manual item handling

Typical transaction time: 5 – 10 minutes
Full customer support

Detailed next

Self-service checkout
15-20%



Customer – operated checkout

Bagging area included, staff monitored

Typical transaction time: 3 – 5 minutes
Semi-automated process - customers have independence with support available

Frictionless checkout
<1%



AI automatic detection

No customer scanning or interaction required, walk-through payment - items recognized instantly, automatic payment

Transaction time: ~30 seconds (fastest method available)
Completely frictionless - zero checkout effort or waiting

SCO kiosks can serve multiple functions, including retail media

Standard



Self-service kiosk with integrated bagging area

Staff-monitored with one clerk covering multiple terminals

Card and cash payments

Proven technology, **dominant market format**

Hybrid



Convertible operation to adapt to store traffic in real-time

Staff-operated or staff-monitored

Combines assisted and self-service benefits while optimizing labor and service efficiency

Retail media



Advertising-enabled checkout with integrated display capability showing promotional content during **transactions and idle periods**
Creates **new revenue stream** for retailers by selling **promotional space to brands** throughout operational hours, while reaching customers at **point of purchase** when highest receptivity to offers with flexible implementation options

Add-on technologies

Handheld scanners

Enable customers to scan products while shopping and carry transactions to checkout station, cost-efficient solution integrating with existing retail infrastructure

Smart carts

Shopping carts with built-in scanners for Scan & Go functionality and real-time basket tracking, complete payment at cart or checkout

RFID

Automatic product detection via RFID tags without barcode scanning

Computer vision AI

3D camera with AI recognition system, fastest SCO technology available

Usage of our SCOs shows smaller baskets which are over-indexed on short-lived food and impulse missions

		SCO vs. Network
Avg. Basket, items		~4,0 ↘
Basket structure, % of sales value	Short-lived food	↗
	Long-lived food	↗
	Fresh top-up	→
	Beverages	↗
	Sweets Snacks Cakes	→
	Alcohols	↘
	Non-food	↘
Missions, % of trx	Impulse	↗
	Emergency	↘
	Top-up	↘
	Fresh	→
	Stock-up	→
	Promo	→



SCO implementation should be set up along two main goals

Format goals

Consistently provide formats, features and assortment to customers in microlocation where they are expected

E.g., Customers in fast-paced, impulse-oriented Urban convenience format are quick to adopt technology and eager for an option of self-paced/self-guided checkout

Self-service checkout is very attractive for small baskets in time-sensitive impulse/emergency missions which are more frequent in urban settings

Operational goals

Enable our store employees to focus on value-adding tasks and services by removing obstacles and providing tools

E.g., Store employees can focus on helpful customer interactions or non-automatizable tasks such as restocking

Self-service checkout helps in stores with strong peak hours transactions which can lead to loss of sales due to long lines



Case study details



Task description



What makes a good SCO candidate?

Define the criteria: transaction volume, basket size, time-of-day patterns, staffing implications, store format.



Develop scoring and metrics

Translate criteria into measurable indicators. Consider how you would weight different factors against each other.



What is your proposed framework?

Define your rules on when should SCO be included, what is the benefit of the proposed solution? What are limitations? What are your assumptions? Etc.

Your goal is not to simply analyse the provided dataset

It is to develop a reusable framework for evaluating store-level SCO potential, then apply and validate it using the supplied data

Dataset notes

Dataset of store locations and half-hourly transaction data is your testing ground, use it to **test your framework**, **illustrate how it works in practice**, and **refine your criteria based** on what the data reveals.



Dataset includes stores both with and without existing SCO installations. However, this should not be treated as a ready-made training set. The absence of SCO in a store may reflect space constraints, capital prioritisation, or rollout sequencing rather than any judgment that the store is a poor candidate. Equally, the presence of SCO does not guarantee it was deployed optimally. Do not assume the current state of deployment is the correct answer, your framework should form its own view, independent of what already exists on the ground.



Transaction dataset is synthetic, generated to reflect the statistical distributions of real store data, but not representative of any actual locations or market conditions. It should be used purely as a vehicle for applying and demonstrating your framework, not as a source of genuine market insight.

Deliverables & Evaluation

Deliverables



Contestants will be judged based on provided materials that showcase their thought process.

1

Case study PDF

Framework, methodology, findings and recommendations in a professional business format

2

Data Exploration

Supporting analysis in your tool of choice: Python, Excel, R, or other.

3

Presentation

Shortlisted teams present to a judging panel and defend their methodology. Only if shortlisted.

Evaluation



Competition is judged on the quality of thinking and methodology, not on arriving at a single correct answer. There is no definitive right result. A simple solution can win if the data is backing it up.

Framework logic

Data exploration

Communication

Creativity & Insight

